

Week 1 • Day 5

Exploratory Data Analysis — Part 2

Analysed all categorical features — internet service, payment methods, and add-on services

DATE	INTERN NAME	PROJECT	MENTOR
Day 5 of 28 Week 1		Customer Churn Prediction & LTV Engine	

■ Technologies Used Today

Python

Pandas

Seaborn

Matplotlib

NumPy

■ Today's Key Metrics

4

New Charts Created

45.3%

Electronic Check Churn

41.9%

Fiber Optic Churn

42%

No-Security Churn Rate

■ Day Objectives

- Analyse churn rates across all categorical features
- Compare churn: Fiber optic vs DSL vs No internet
- Identify highest-churn payment method
- Visualise impact of 6 add-on services on churn rate
- Analyse senior citizen churn vs non-senior

■ Tasks Completed Today

1. Created notebooks/eda_part2.py — categorical EDA script

2. Chart 6: Internet service churn — count plot + churn rate bar (Fiber optic 41.9%, DSL 19.0%, No 7.4%)
3. Chart 7: Payment method churn — sorted bar chart (Electronic check highest at 45.3%)
4. Chart 8: Add-on services comparison — grouped bar chart showing With vs Without each service
5. Chart 9: Senior citizen churn — 41.7% senior vs 23.7% non-senior
6. Printed summary report to terminal with all key churn rates
7. Documented that OnlineSecurity absence = 42% churn vs 14.6% with it
8. Documented that auto-payment methods reduce churn to ~15%
9. Saved all 4 charts (06-09) to reports/figures/
10. Added annotations (percentage labels) to all bar charts
11. Committed all files to GitHub

■ Concepts Learned

- Why Fiber optic customers churn more — likely due to higher price point
- Electronic check as a churn risk indicator — possibly correlated with less engaged customers
- Protective effect of add-on services — bundle customers are more sticky
- How to create grouped bar charts with `numpy.arange` for positioning
- `pandas.groupby + lambda` apply pattern for computing churn rates by group
- Why senior citizens may churn more — possibly price sensitivity on fixed incomes

■ Tools / Software Used

Tool / Library	Version / Notes	Status
Seaborn	0.13.0	✓ In use
Matplotlib	3.8.2	✓ In use
NumPy	1.26.2	✓ In use

■ Outputs / Deliverables Produced

- ◆ reports/figures/06_internet_service_churn.png
- ◆ reports/figures/07_payment_method_churn.png
- ◆ reports/figures/08_addon_services_churn.png
- ◆ reports/figures/09_senior_citizen_churn.png
- ◆ notebooks/eda_part2.py — categorical analysis script
- ◆ Terminal summary report with all churn rates
- ◆ GitHub commit with 4 new charts

■ Problems Encountered & Solutions

Problem Faced	How I Solved It
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Bar chart x-axis labels overlapping for payment methods	Added <code>plt.xticks(rotation=15, ha='right')</code> to rotate and right-align labels
Grouped bar chart bars not centred correctly	Used <code>numpy.arange()</code> and width offsets ($x-w/2$ and $x+w/2$) to position grouped bars

■ GitHub Commit Record

Commit Message	Week1-Day5: EDA Part 2 — categorical analysis, 4 additional charts, payment/service insights
Branch	main
Files Changed	notebooks/eda_part2.py, reports/figures/06-09 PNG charts
Commit Hash	(fill after push)

■ End-of-Day Submission Checklist

■ 4 new charts saved (06-09)	■ Fiber optic churn = 41.9%
■ Electronic check = 45.3% churn	■ Add-on services chart done
■ Senior citizen chart done	■ All bar charts annotated
■ Summary printed to terminal	■ 9 total charts in reports/
■ eda_part2.py committed	■ Day 5 commit on GitHub

■ Self Assessment

9/10 Self Rating	6 Hours Spent	Week 1, Day 5 Progress	5/28 Day in Program
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■ Key Learnings & Reflections

- ★ Customers without OnlineSecurity churn at 42% vs 14.6% with it — strongest protective signal
- ★ Electronic check users churn 3× more than credit card auto-pay users
- ★ Every add-on service reduces churn — bundle strategy is key recommendation

★ These categorical insights directly inform Week 2 feature engineering decisions

■ Plan for Tomorrow

- Identify and fix all data quality issues in the dataset
- Fix TotalCharges 11 empty strings by imputing with MonthlyCharges
- Standardise column names to lowercase
- Convert Yes/No binary columns to 1/0 for machine learning
- Save cleaned dataset to data/processed/

INTERN SIGNATURE	MENTOR SIGNATURE	DATE SUBMITTED